Surface Metrics Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Text Descriptives on the Exchange-Aligned Structure

PENPAL Project

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Contents

1 Overview

This notebook compares surface-level text metrics across conditions using the long-format, exchange-aligned metric tables.

The underlying processed data are now one row per **speaker contribution**, with recovered role metadata:

- **Human / AI** in Human-AI
- **Starter-side / Non-starter-side** in Human-Human and AI-AI

Those role labels are useful for **within-condition description**, but they are not the primary cross-condition comparison axis. For cross-condition comparison, this notebook uses **Slot 1 / Slot 2 asymmetry**.

The notebook therefore no longer assumes old wide `author_1_*` / `author_2_*` columns.

Primary cross-condition object. Per story and per metric m,

$$A_m(i) \;=\; \left|\,\overline{m}_{\text{slot}=1,\,i} - \overline{m}_{\text{slot}=2,\,i}\,\right|.$$

This is the unsigned, label-invariant side-difference magnitude. For each metric we test $A_m \sim$ condition with ANOVA + Kruskal and apply BH adjustment across the metric panel. The LMM slot-marginal contrasts are retained as supplemental. Starter-mechanism and Human-AI speaker-type decompositions are deferred.

2 Load Data

Table 1: Loaded surface-metric rows by condition

condition	n_stories	n_rows
Human-AI	97	1744
Human-Human	31	558
AI-AI	40	720

3 Metric Selection

Table 2: Surface metrics selected for the comparison notebook

metric_alias	column_name
word_count	n_tokens
sentence_length	sentence_length_mean
lexical_diversity	proportion_unique_tokens
readability	flesch_reading_ease
coherence	first_order_coherence

4 Primary: Per-Story Side-Mean Difference

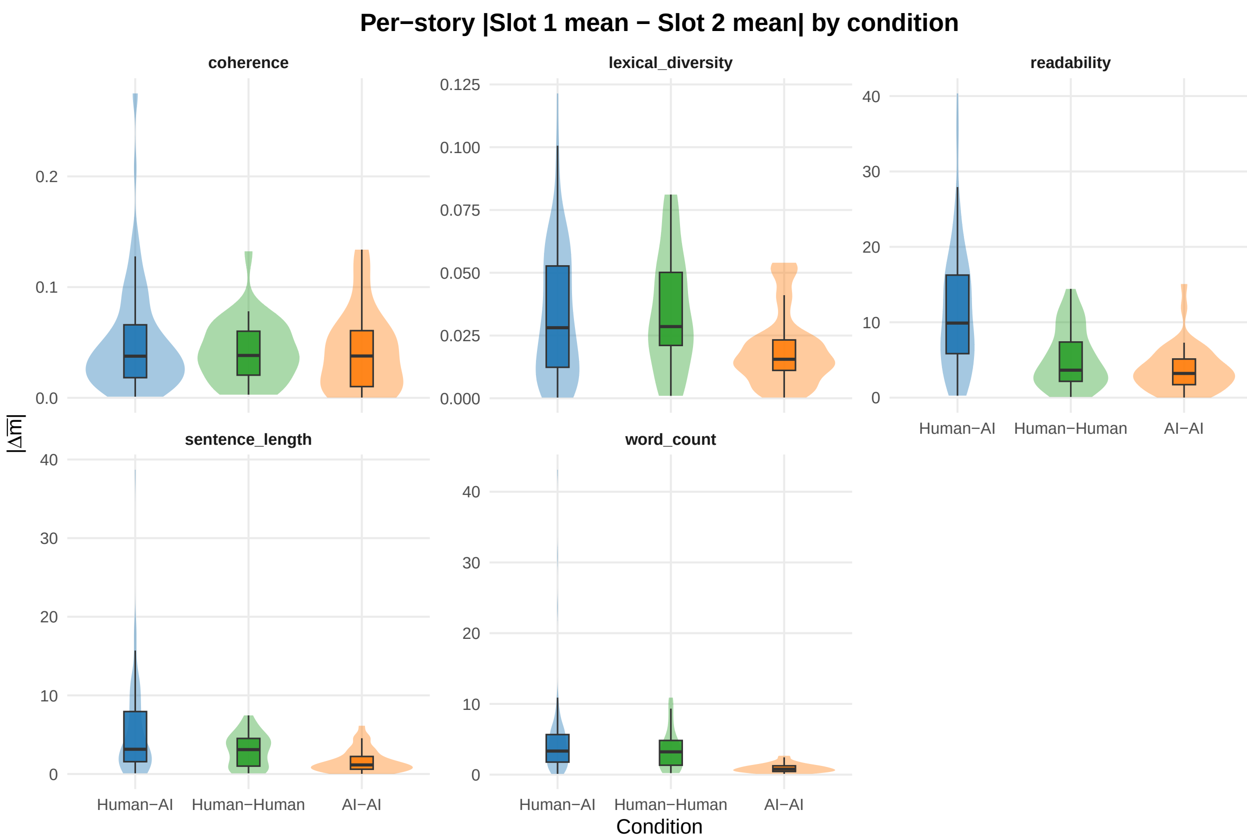
Table 3: Per-story |side-mean difference| by condition, per metric

metric	condition	n	mean_A	median_A	sd_A
coherence	Human-AI	95	0.051	0.038	0.051
coherence	Human-Human	31	0.042	0.038	0.028
coherence	AI-AI	39	0.042	0.038	0.037
lexical_diversity	Human-AI	97	0.033	0.028	0.026
lexical_diversity	Human-Human	31	0.035	0.029	0.023
lexical_diversity	AI-AI	40	0.019	0.016	0.015
readability	Human-AI	97	11.415	9.893	8.046
readability	Human-Human	31	5.159	3.644	3.893
readability	AI-AI	40	3.793	3.210	3.086
sentence_length	Human-AI	97	5.295	3.148	5.742
sentence_length	Human-Human	31	3.023	3.102	2.105
sentence_length	AI-AI	40	1.661	1.157	1.505
word_count	Human-AI	97	4.719	3.333	5.795
word_count	Human-Human	31	3.599	3.222	2.931
word_count	AI-AI	40	0.944	0.778	0.620

Table 4: Primary cross-condition tests for A_m ~ condition, per metric (BH-adjusted across metrics)

metric	anova_p	kw_p	anova_p_BH	kw_p_BH
coherence	0.3960	0.699	0.3960	0.6990
lexical_diversity	0.0038	0.005	0.0048	0.0063

metric	anova_p	kw_p	anova_p_BH	kw_p_BH
readability	0.0000	0.000	0.0000	0.0000
sentence_length	0.0001	0.000	0.0002	0.0001
word_count	0.0001	0.000	0.0002	0.0000

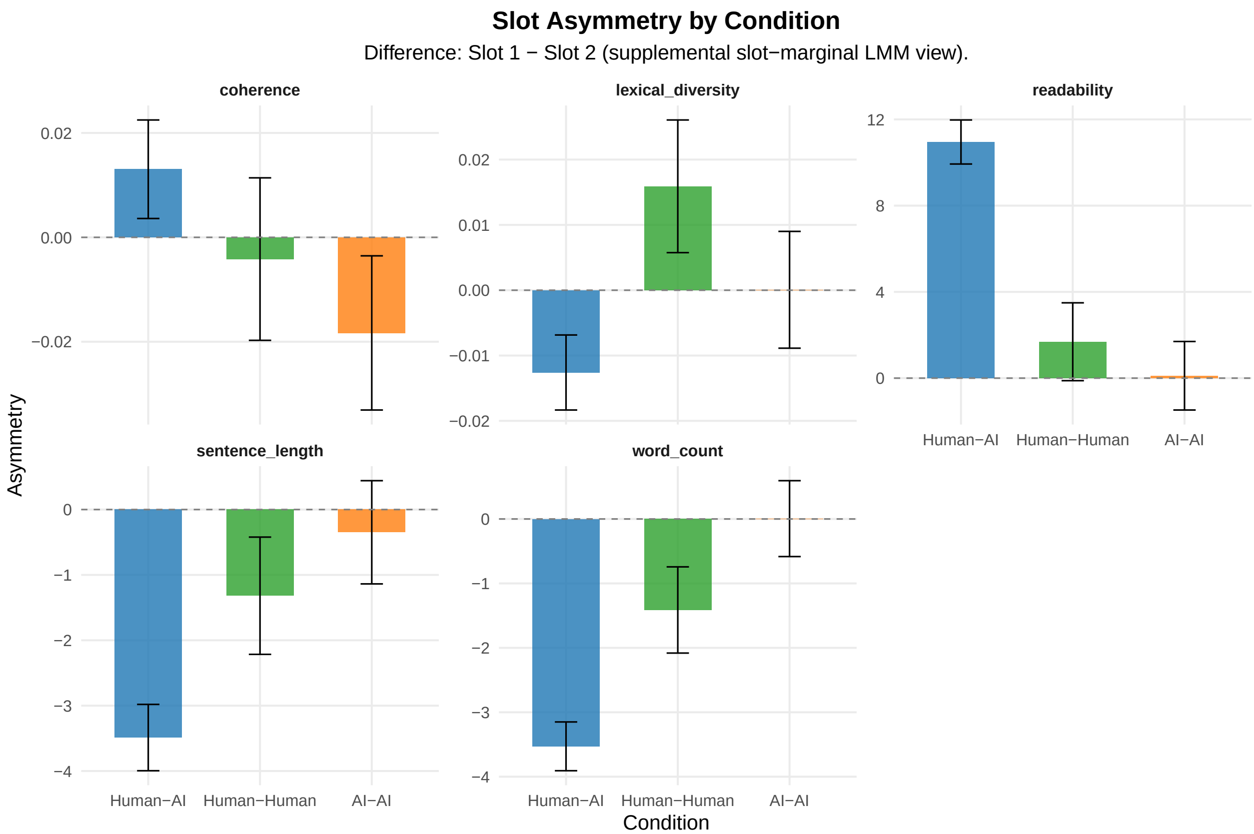


5 Supplemental: Slot-Marginal LMM per Metric

Table 5: Slot asymmetry by condition for selected surface metrics

metric	condition	slot_1_mean	slot_2_mean	difference	se	p	t
word_count	Human-AI	25.602	29.130	-3.529	0.193	0.000	NA
word_count	Human-Human	25.975	27.387	-1.412	0.341	0.000	NA
word_count	AI-AI	27.972	27.967	0.006	0.301	0.985	NA
sentence_length	Human-AI	11.457	14.945	-3.488	0.259	0.000	NA
sentence_length	Human-Human	10.000	11.319	-1.319	0.458	0.004	NA
sentence_length	AI-AI	14.904	15.252	-0.348	0.403	0.388	NA
lexical_diversity	Human-AI	0.870	0.883	-0.013	0.003	0.000	NA
lexical_diversity	Human-Human	0.873	0.857	0.016	0.005	0.002	NA
lexical_diversity	AI-AI	0.879	0.879	0.000	0.005	0.988	NA
readability	Human-AI	90.868	79.916	10.952	0.521	0.000	NA

metric	condition	slot_1_mean	slot_2_mean	difference	se	p	t
readability	Human-Human	94.081	92.393	1.688	0.921	0.067	NA
readability	AI-AI	83.061	82.951	0.110	0.811	0.892	NA
coherence	Human-AI	0.804	0.791	0.013	0.005	0.007	2.709
coherence	Human-Human	0.816	0.820	-0.004	0.008	0.600	-0.525
coherence	AI-AI	0.777	0.795	-0.018	0.008	0.015	-2.430

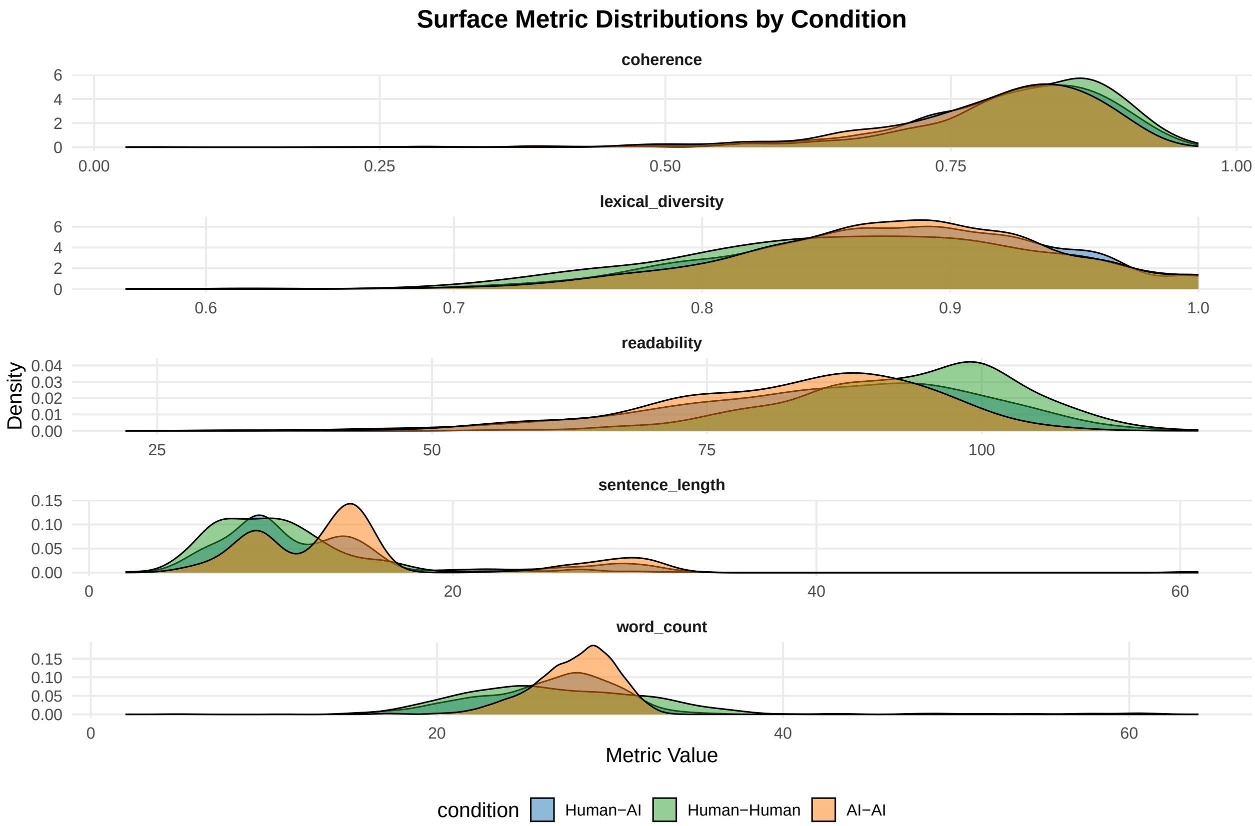


6 Supplemental: General Condition-Level Comparison

Table 6: Condition-level descriptives for selected surface metrics

condition	metric	mean_value	sd_value
Human-AI	word_count	27.365	6.350
Human-Human	word_count	26.681	4.848
AI-AI	word_count	27.969	2.409
Human-AI	sentence_length	13.202	7.151
Human-Human	sentence_length	10.660	4.521
AI-AI	sentence_length	15.078	6.987
Human-AI	lexical_diversity	0.876	0.065
Human-Human	lexical_diversity	0.865	0.071
AI-AI	lexical_diversity	0.879	0.061
Human-AI	readability	85.397	14.411

condition	metric	mean_value	sd_value
Human-Human	readability	93.237	10.314
AI-AI	readability	83.006	12.020
Human-AI	coherence	0.798	0.099
Human-Human	coherence	0.818	0.079
AI-AI	coherence	0.785	0.099




```
##
## === word_count ( n_tokens ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
##    Data: df_combined
##
## REML criterion at convergence: 17760.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -6.5161 -0.5153  0.0778  0.5060  5.4540
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 10.94    3.307
##  Residual                    18.24    4.271
## Number of obs: 3022, groups:  conversation_id, 168
##
```

```
## Fixed effects:
##                      Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)           27.3661     0.3510 165.0340  77.960   <2e-16 ***
## conditionHuman-Human  -0.6851     0.7133 165.0072  -0.961    0.338
## conditionAI-AI         0.6033     0.6496 165.0090   0.929    0.354
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) cndH-H
## cndtnHmn-Hm -0.492
## condtnAI-AI -0.540  0.266

##  contrast                    estimate    SE  df z.ratio p.value
##  (Human-AI) - (Human-Human)    0.685 0.713 Inf   0.961  1.0000
##  (Human-AI) - (AI-AI)         -0.603 0.650 Inf  -0.929  1.0000
##  (Human-Human) - (AI-AI)      -1.288 0.827 Inf  -1.558  0.3580
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === sentence_length ( sentence_length_mean ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
##    Data: df_combined
##
## REML criterion at convergence: 19331.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.9998 -0.6040 -0.1445  0.3940  5.7394
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 14.03    3.745
##  Residual                    31.12    5.579
## Number of obs: 3022, groups:  conversation_id, 168
##
## Fixed effects:
##                      Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)           13.2006     0.4030 165.0418  32.752  < 2e-16 ***
## conditionHuman-Human  -2.5411     0.8190 165.0074  -3.103  0.00226 **
## conditionAI-AI         1.8773     0.7459 165.0097   2.517  0.01279 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) cndH-H
## cndtnHmn-Hm -0.492
## condtnAI-AI -0.540  0.266

##  contrast                    estimate    SE  df z.ratio p.value
##  (Human-AI) - (Human-Human)     2.54 0.819 Inf   3.103  0.0058
##  (Human-AI) - (AI-AI)          -1.88 0.746 Inf  -2.517  0.0355
```

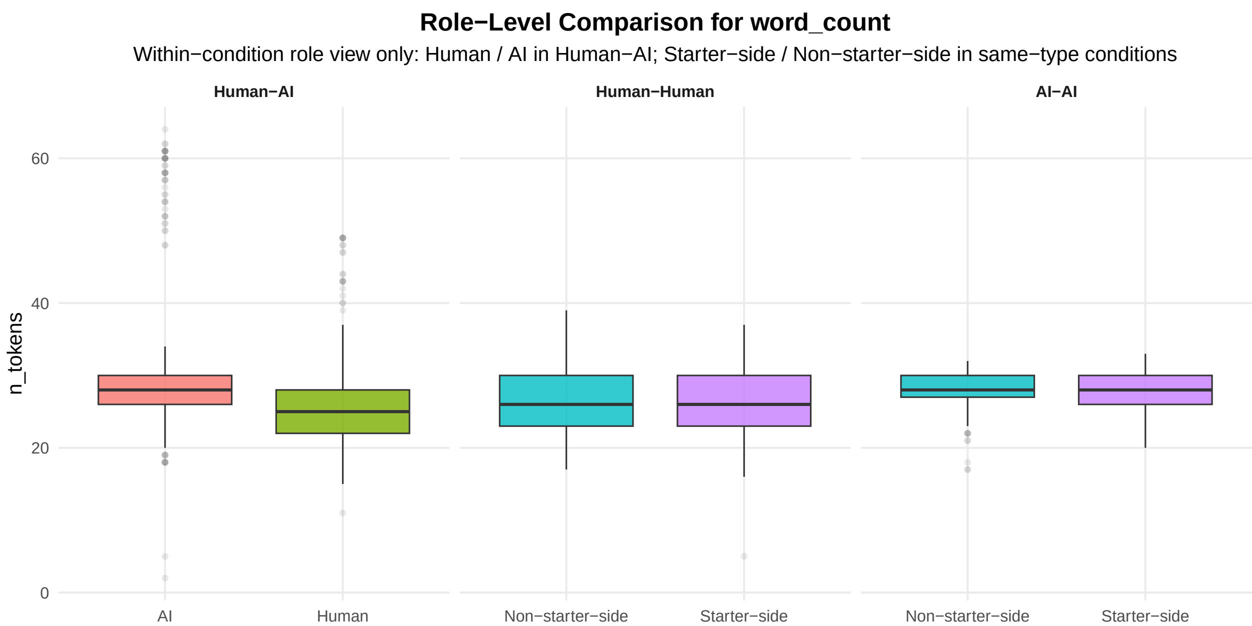
```
##  (Human-Human) - (AI-AI)       -4.42 0.950 Inf  -4.652  <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === lexical_diversity ( proportion_unique_tokens ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
##    Data: df_combined
##
## REML criterion at convergence: -8053.9
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.6064 -0.6339  0.0442  0.6826  2.5006
##
## Random effects:
##  Groups          Name        Variance  Std.Dev.
##  conversation_id (Intercept) 0.0004996 0.02235
##  Residual                    0.0037766 0.06145
## Number of obs: 3022, groups:  conversation_id, 168
##
## Fixed effects:
##                        Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)            0.876246   0.002705 164.902751 323.965   <2e-16 ***
## conditionHuman-Human  -0.011100   0.005495 164.814136  -2.020    0.045 *
## conditionAI-AI         0.003149   0.005005 164.819956   0.629    0.530
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) cndH-H
## cndtnHmn-Hm -0.492
## condtnAI-AI -0.540  0.266
##  contrast                    estimate      SE  df z.ratio p.value
##  (Human-AI) - (Human-Human)  0.01110 0.00550 Inf   2.020  0.1302
##  (Human-AI) - (AI-AI)       -0.00315 0.00500 Inf  -0.629  1.0000
##  (Human-Human) - (AI-AI)    -0.01425 0.00637 Inf  -2.236  0.0761
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === readability ( flesch_reading_ease ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
##    Data: df_combined
##
## REML criterion at convergence: 23729.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -4.5708 -0.6114  0.0903  0.6798  2.8224
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept)  37.88    6.154
##  Residual                    136.69   11.691
## Number of obs: 3022, groups:  conversation_id, 168
##
## Fixed effects:
##                      Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)           85.3924     0.6847 165.0437 124.710  < 2e-16 ***
## conditionHuman-Human   7.8445     1.3913 164.9920   5.638 7.29e-08 ***
## conditionAI-AI        -2.3866     1.2671 164.9954  -1.883   0.0614 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) cndH-H
## cndtnHmn-Hm -0.492
## condtnAI-AI -0.540  0.266

##  contrast                    estimate   SE  df z.ratio p.value
##  (Human-AI) - (Human-Human)    -7.84 1.39 Inf  -5.638  <.0001
##  (Human-AI) - (AI-AI)           2.39 1.27 Inf   1.883  0.1789
##  (Human-Human) - (AI-AI)       10.23 1.61 Inf   6.341  <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === coherence ( first_order_coherence ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
##    Data: df_combined
##
## REML criterion at convergence: -4885.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -7.8824 -0.4498  0.1673  0.6560  1.8889
##
## Random effects:
##  Groups          Name        Variance  Std.Dev.
##  conversation_id (Intercept) 0.0007412 0.02722
##  Residual                    0.0083634 0.09145
## Number of obs: 2596, groups:  conversation_id, 168
##
## Fixed effects:
##                        Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)            0.798403   0.003673 162.569767 217.350  < 2e-16 ***
## conditionHuman-Human   0.019474   0.007295 151.227816   2.670  0.00842 **
## conditionAI-AI        -0.012731   0.006891 163.760226  -1.848  0.06647 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation of Fixed Effects:
##              (Intr) cndH-H
## cndtnHmn-Hm -0.504
## condtnAI-AI -0.533  0.268
##  contrast                    estimate      SE  df t.ratio p.value
##  (Human-AI) - (Human-Human)  -0.0195 0.00729 153  -2.670  0.0252
##  (Human-AI) - (AI-AI)         0.0127 0.00689 166   1.847  0.1995
##  (Human-Human) - (AI-AI)      0.0322 0.00859 157   3.751  0.0007
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
```

7 Supplemental: Role-Level Comparison




```
##
## === Role contrast for word_count ===
##
## Human-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##    Data: df_subset
##
## REML criterion at convergence: 10494.9
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -5.7918 -0.4917  0.0097  0.4773  4.6860
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
```

```
##  conversation_id (Intercept) 16.71    4.088
##  Residual                    20.65    4.544
## Number of obs: 1744, groups:  conversation_id, 97
##
## Fixed effects:
##                       Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)            29.1305     0.4427  108.7364   65.81   <2e-16 ***
## condition_roleHuman    -3.5287     0.2176 1646.0151  -16.21   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## cndtn_rlHmn -0.246
##  contrast   estimate    SE   df t.ratio p.value
##  AI - Human     3.53 0.218 1646  16.214  <.0001
##
## Degrees-of-freedom method: kenward-roger
##
## Human-Human
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##    Data: df_subset
##
## REML criterion at convergence: 3260.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.2688 -0.7333 -0.0420  0.7141  2.6563
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept)  5.408   2.326
##  Residual                    18.288   4.276
## Number of obs: 558, groups:  conversation_id, 31
##
## Fixed effects:
##                             Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                  26.7742     0.4899  40.1823  54.652   <2e-16 ***
## condition_roleStarter-side  -0.1864     0.3621 526.0000  -0.515    0.607
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## cndtn_rlSt- -0.370
##  contrast                            estimate    SE  df t.ratio p.value
##  (Non-starter-side) - (Starter-side)    0.186 0.362 526   0.515  0.6069
##
## Degrees-of-freedom method: kenward-roger
##
## AI-AI
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##    Data: df_subset
##
## REML criterion at convergence: 3191.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.9283 -0.5796  0.1691  0.6762  2.4627
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 1.419    1.191
##  Residual                    4.424    2.103
## Number of obs: 720, groups:  conversation_id, 40
##
## Fixed effects:
##                             Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)                 28.00278    0.21854  51.30301 128.136   <2e-16 ***
## condition_roleStarter-side  -0.06667    0.15678 679.00000  -0.425    0.671
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## cndtn_rlSt- -0.359
##  contrast                            estimate    SE  df t.ratio p.value
##  (Non-starter-side) - (Starter-side)   0.0667 0.157 679   0.425  0.6708
##
## Degrees-of-freedom method: kenward-roger
##
## === Role contrast for sentence_length ===
##
## Human-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##    Data: df_subset
##
## REML criterion at convergence: 11298.4
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.5195 -0.6081 -0.1773  0.3810  5.2226
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 14.41    3.796
##  Residual                    33.83    5.816
## Number of obs: 1744, groups:  conversation_id, 97
##
```

```
## Fixed effects:
##                      Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)           14.9446     0.4328  119.3930   34.53   <2e-16 ***
## condition_roleHuman   -3.4879     0.2785 1646.0245  -12.52   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## cndtn_rlHmn -0.322
##  contrast   estimate    SE   df t.ratio p.value
##  AI - Human     3.49 0.279 1646  12.522  <.0001
##
## Degrees-of-freedom method: kenward-roger
##
## Human-Human
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##    Data: df_subset
##
## REML criterion at convergence: 3258.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.6805 -0.6623 -0.1925  0.3109  5.2171
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept)  0.7956  0.892
##  Residual                    19.5527  4.422
## Number of obs: 558, groups:  conversation_id, 31
##
## Fixed effects:
##                            Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                 11.0514     0.3094  73.2373  35.715   <2e-16 ***
## condition_roleStarter-side  -0.7836     0.3744 526.0000  -2.093   0.0368 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## cndtn_rlSt- -0.605
##  contrast                            estimate    SE  df t.ratio p.value
##  (Non-starter-side) - (Starter-side)    0.784 0.374 526   2.093  0.0368
##
## Degrees-of-freedom method: kenward-roger
##
## AI-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
```

```
##    Data: df_subset
##
## REML criterion at convergence: 4487.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.2291 -0.7053  0.0685  0.5080  3.4982
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 23.70    4.868
##  Residual                    25.52    5.052
## Number of obs: 720, groups:  conversation_id, 40
##
## Fixed effects:
##                            Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                 15.5219     0.8145  43.5195  19.057   <2e-16 ***
## condition_roleStarter-side  -0.8880     0.3766 679.0000  -2.358   0.0186 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## cndtn_rlSt- -0.231
##  contrast                            estimate    SE  df t.ratio p.value
##  (Non-starter-side) - (Starter-side)    0.888 0.377 679   2.358  0.0186
##
## Degrees-of-freedom method: kenward-roger
##
## === Role contrast for lexical_diversity ===
##
## Human-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##    Data: df_subset
##
## REML criterion at convergence: -4653.8
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.9751 -0.6353  0.0508  0.7071  2.5783
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 0.000426 0.02064
##  Residual                    0.003779 0.06148
## Number of obs: 1744, groups:  conversation_id, 97
##
## Fixed effects:
##                       Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)          8.825e-01  2.954e-03  1.685e+02 298.760  < 2e-16 ***
## condition_roleHuman -1.260e-02  2.944e-03  1.646e+03  -4.278 1.99e-05 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## cndtn_rlHmn -0.498
##  contrast   estimate      SE   df t.ratio p.value
##  AI - Human   0.0126 0.00294 1646   4.278  <.0001
##
## Degrees-of-freedom method: kenward-roger
##
## Human-Human
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##    Data: df_subset
##
## REML criterion at convergence: -1366.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.1590 -0.6518  0.0436  0.7255  2.1796
##
## Random effects:
##  Groups          Name        Variance  Std.Dev.
##  conversation_id (Intercept) 0.0004418 0.02102
##  Residual                    0.0046542 0.06822
## Number of obs: 558, groups:  conversation_id, 31
##
## Fixed effects:
##                              Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)                  0.866006   0.005562  55.807606 155.710   <2e-16
## condition_roleStarter-side  -0.001720   0.005776 526.000000  -0.298    0.766
##
## (Intercept)                ***
## condition_roleStarter-side
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## cndtn_rlSt- -0.519
##  contrast                              estimate      SE  df t.ratio p.value
##  (Non-starter-side) - (Starter-side)  0.00172 0.00578 526   0.298  0.7660
##
## Degrees-of-freedom method: kenward-roger
##
## AI-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##    Data: df_subset
```

```
##
## REML criterion at convergence: -2055.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.96440 -0.64985  0.03278  0.63825  2.75075
##
## Random effects:
##  Groups          Name        Variance  Std.Dev.
##  conversation_id (Intercept) 0.0007296 0.02701
##  Residual                    0.0030035 0.05480
## Number of obs: 720, groups:  conversation_id, 40
##
## Fixed effects:
##                             Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)                8.785e-01  5.156e-03 5.476e+01  170.38   <2e-16 ***
## condition_roleStarter-side 1.798e-03  4.085e-03 6.790e+02    0.44     0.66
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## cndtn_rlSt- -0.396
##  contrast                            estimate      SE  df t.ratio p.value
##  (Non-starter-side) - (Starter-side)  -0.0018 0.00408 679  -0.440  0.6599
##
## Degrees-of-freedom method: kenward-roger
```

8 Summary

Primary cross-condition result: per-metric $A_m(i) = |\overline{m}_{\text{slot }1,i} - \overline{m}_{\text{slot }2,i}|$, tested with ANOVA + Kruskal and BH-adjusted across metrics.

Supplemental views retained:

- LMM slot-marginal contrasts per metric (`metric ~ slot * condition + (1|story)`)
- condition-level distributions and marginal `condition` LMMs
- within-condition role comparisons (Human / AI in Human-AI; Starter-side / Non-starter-side in same-type)

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized side differences across all conditions)
- Human-AI decomposition (`speaker_type \u00d7 speaker_is_starter` within Human-AI only)

Analysis completed: 2026-04-23 12:20:00.127788